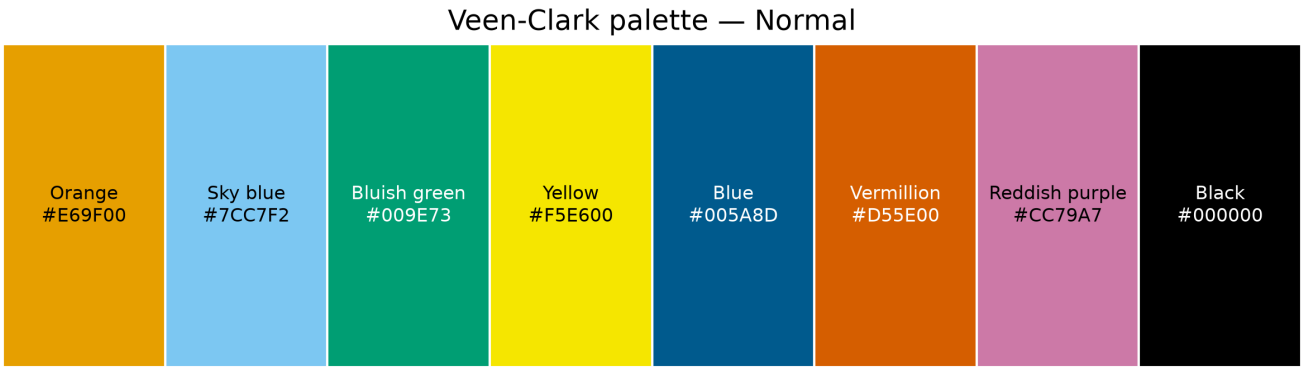
# Veen-Clark palette

## Version 0.1.0 evaluation

Archive DOI: 10.5281/zenodo.23418261




Veen-Clark palette in normal sRGB.

An Okabe-Ito derivative by Roland J. Veen and Charlie J. Clark. Sky blue is lightened and blue is darkened so the pair can represent related subgroups in scientific scatter and bar charts. The palette uses yellow #F5E600.

This report compares colour-difference metrics for the Veen-Clark and Okabe-Ito palettes.

Generated with PaletteBench 0.1.0 using Python 3.11.16. Exact inputs, hashes, dependencies, full-precision CSV/JSON data, and individual vector figures accompany this PDF in the repository.

# Results

The intended sky-blue/blue subgroup pair has greater modelled separation relative to Okabe-Ito under every standard reported condition: normal, protan 100%, deutan 100%, tritan 100%, and grayscale.

| Condition | Okabe-Ito | Veen-Clark | Change |
|---|---|---|---|
| Normal | 22.31 | 37.92 | +15.61 |
| Protan 100% | 20.75 | 36.03 | +15.28 |
| Deutan 100% | 22.45 | 38.57 | +16.12 |
| Tritan 100% | 19.95 | 35.08 | +15.13 |
| Grayscale | 21.62 | 37.43 | +15.81 |

The whole-palette result is mixed. The normal-vision 10th percentile rises by 2.73 Delta E00 and the protan-100% minimum rises by 3.56, but darkening blue reduces separation from black and produces the largest loss against vermillion under tritan 100% (-12.20 Delta E00).

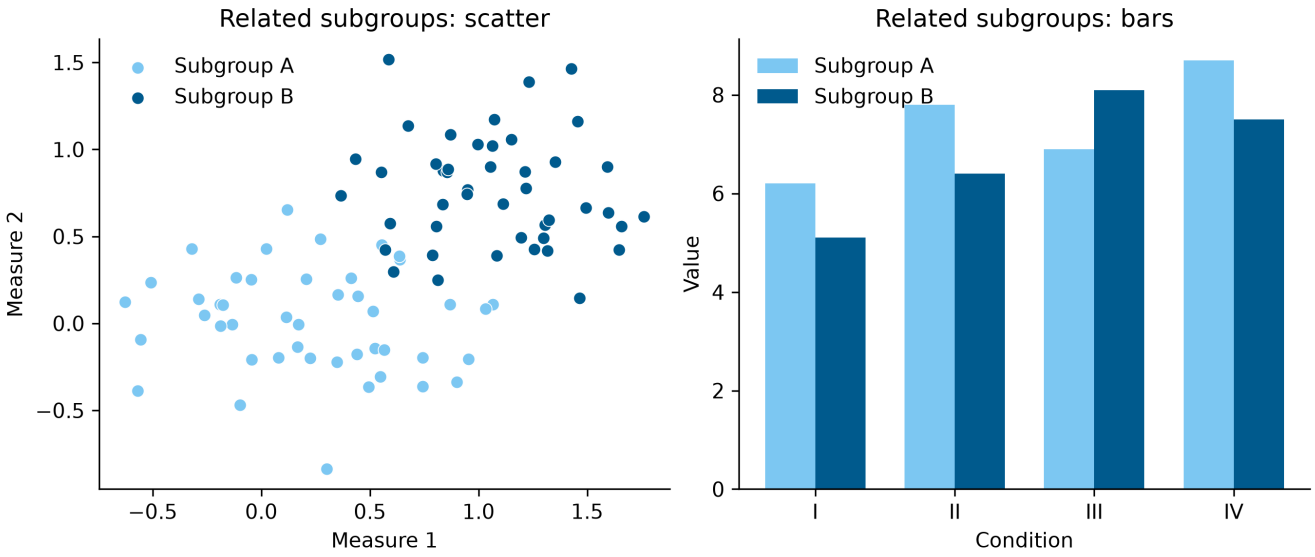



Figure 1. Synthetic data illustrating the two blues with circular markers and solid bar fills.

# Yellow in context

Informal feedback from people with regular colour vision suggested that the original yellow could appear greenish and insufficiently distinct in presentations. This motivated the yellow choice; it was not a controlled participant study. The table reports modelled separation from orange and bluish green.

| Condition | Pair | Okabe-Ito | Veen-Clark | Change |
|---|---|---|---|---|
| Normal | Yellow / orange | 21.73 | 22.04 | +0.31 |
| Normal | Yellow / bluish green | 38.04 | 39.94 | +1.90 |
| Protan 100% | Yellow / orange | 15.17 | 15.81 | +0.64 |
| Protan 100% | Yellow / bluish green | 28.05 | 28.93 | +0.89 |
| Deutan 100% | Yellow / orange | 11.61 | 12.30 | +0.70 |
| Deutan 100% | Yellow / bluish green | 33.90 | 35.47 | +1.58 |
| Tritan 100% | Yellow / orange | 18.83 | 19.15 | +0.32 |
| Tritan 100% | Yellow / bluish green | 43.57 | 43.25 | -0.33 |
| Grayscale | Yellow / orange | 12.83 | 13.29 | +0.47 |
| Grayscale | Yellow / bluish green | 23.32 | 23.78 | +0.46 |

Yellow's normal-sRGB contrast ratio against white is 1.30:1. Pairwise colour separation does not establish visibility on a white slide. PaletteBench does not model projector colour errors or audience perception. Check the actual figure under its intended viewing conditions.

# Baseline-relative comparison

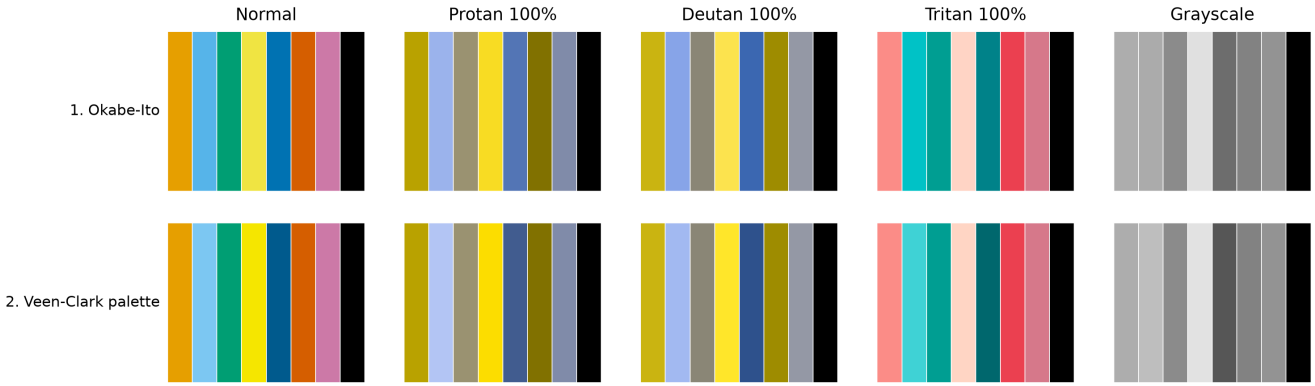



Figure 2. Okabe-Ito baseline and Veen-Clark under the standard reported conditions.

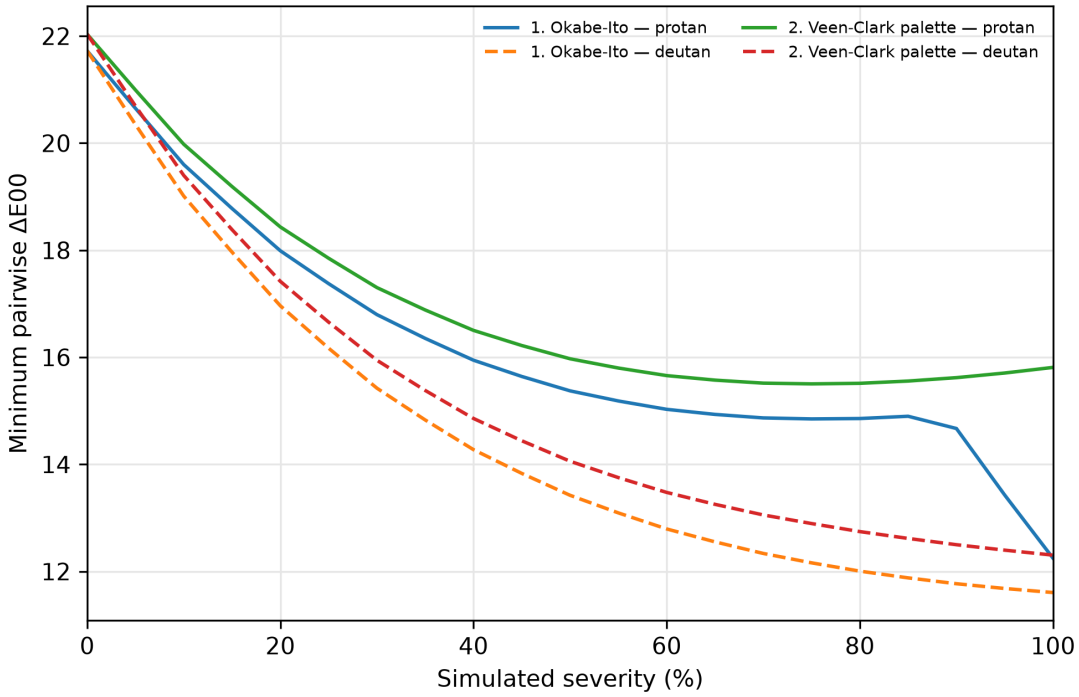



Figure 3. Whole-palette minimum Delta E00 by condition.

# Baseline-relative comparison

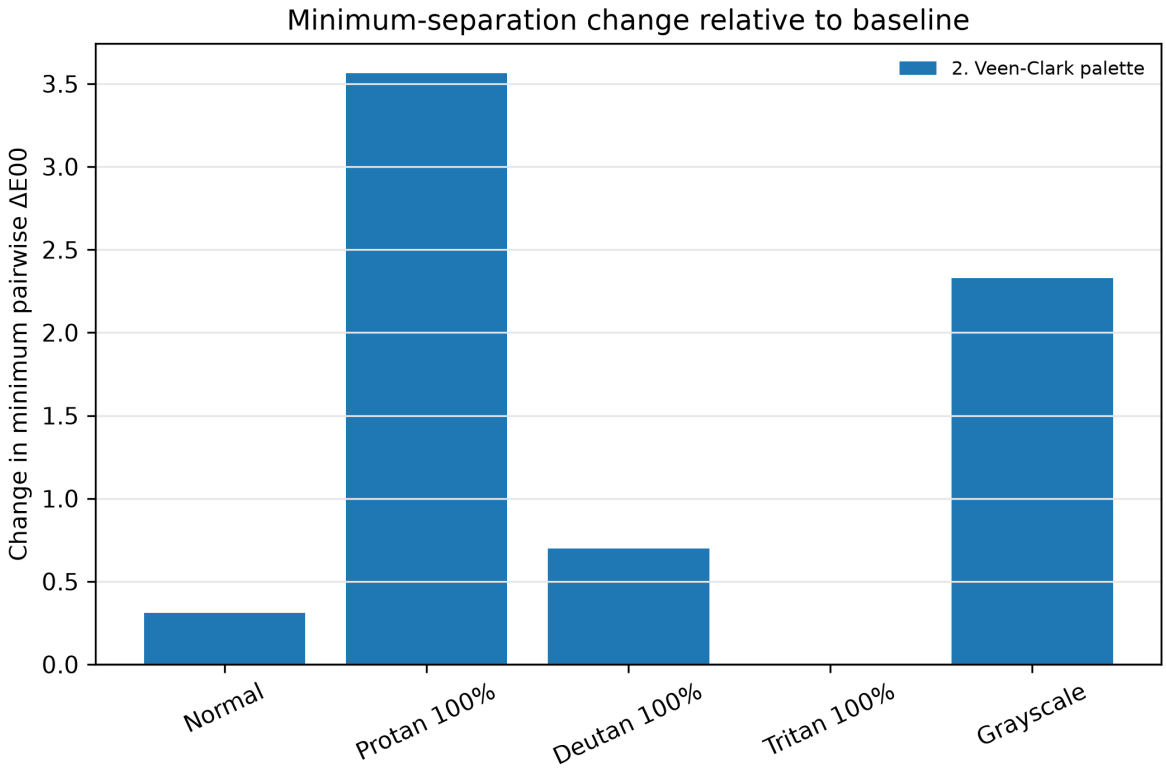



Figure 4. Change in minimum separation relative to Okabe-Ito.

Positive changes indicate greater modelled separation; negative changes indicate reduced modelled separation. Zero change means the minimum separation is unchanged.

# Methods and assumptions

Six-digit values are interpreted as gamma-encoded sRGB. PaletteBench applies Colorspacious sRGB1+CVD simulations based on Machado, Oliveira and Fernandes, clips results to displayable sRGB, converts through CIE XYZ to CIE Lab under the sRGB D65 white point, and calculates CIEDE2000 colour differences with Colour Science for Python. Grayscale preserves WCAG relative luminance.

Severity percentages are simulation parameters, not clinical measurements. Partial tritan severity is omitted from the standard report. Clipping provenance and raw values are retained in the machine-readable audit directory.

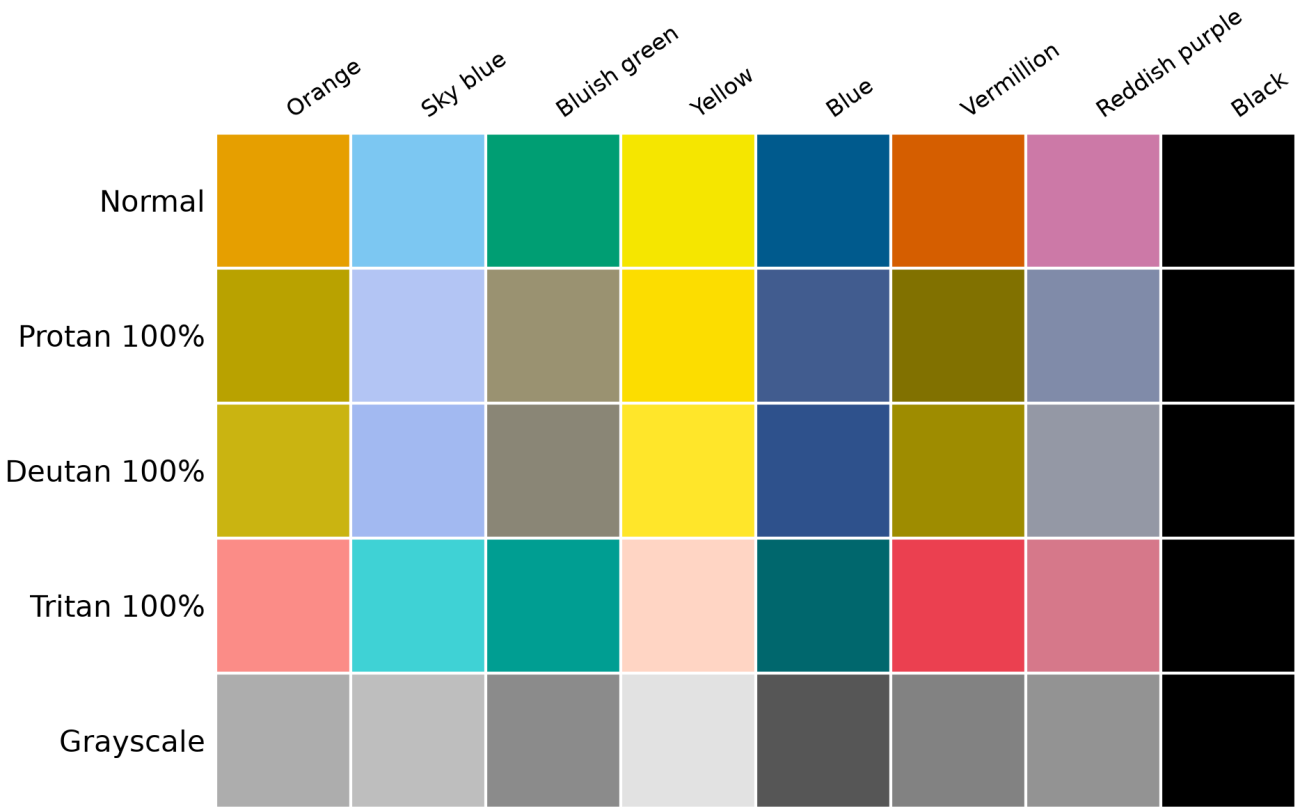



Figure 5. Normal, simulated CVD, and grayscale overview.

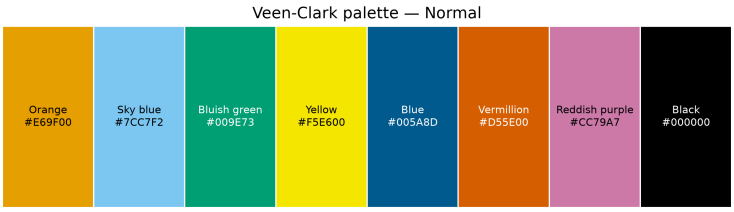
# Palette characteristics

| Colour | Hex | RGB | L* | Chroma | Luminance | Group |
|---|---|---|---|---|---|---|
| Orange | #E69F00 | 230 159 0 | 70.61 | 76.48 | 0.4162 | - |
| Sky blue | #7CC7F2 | 124 199 242 | 77.01 | 30.81 | 0.5154 | blue_family |
| Bluish green | #009E73 | 0 158 115 | 57.74 | 47.49 | 0.2569 | - |
| Yellow | #F5E600 | 245 230 0 | 89.86 | 89.87 | 0.7601 | - |
| Blue | #005A8D | 0 90 141 | 36.43 | 34.62 | 0.0924 | blue_family |
| Vermillion | #D55E00 | 213 94 0 | 54.19 | 76.67 | 0.2215 | - |
| Reddish purple | #CC79A7 | 204 121 167 | 61.05 | 39.97 | 0.2930 | - |
| Black | #000000 | 0 0 0 | 0.00 | 0.00 | 0.0000 | - |

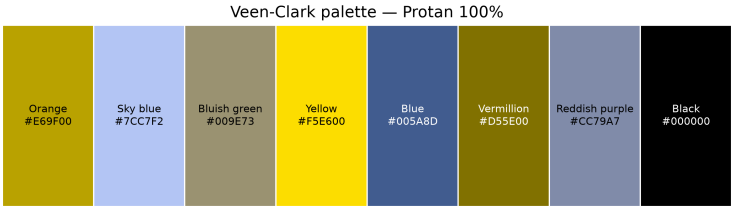
## Condition-level summary

| Condition | Min | Weakest pair | Mean | Median | 10th pct. |
|---|---|---|---|---|---|
| normal | 22.04 | orange / yellow | 49.68 | 49.65 | 33.71 |
| protan20 | 18.43 | orange / yellow | 46.97 | 47.83 | 33.91 |
| protan40 | 16.50 | orange / yellow | 45.05 | 44.26 | 28.67 |
| protan60 | 15.66 | orange / yellow | 43.14 | 42.95 | 23.52 |
| protan80 | 15.51 | orange / yellow | 41.88 | 42.61 | 19.46 |
| protan100 | 15.81 | orange / yellow | 41.32 | 41.46 | 18.08 |
| deutan20 | 17.42 | orange / yellow | 46.89 | 47.93 | 33.41 |
| deutan40 | 14.86 | orange / yellow | 45.12 | 42.83 | 29.74 |
| deutan60 | 13.48 | orange / yellow | 43.37 | 40.27 | 23.95 |
| deutan80 | 12.74 | orange / yellow | 42.23 | 38.78 | 20.38 |
| deutan100 | 12.30 | orange / yellow | 41.70 | 37.53 | 16.01 |
| tritan100 | 10.87 | orange / reddish_purple | 44.53 | 52.05 | 15.79 |
| grayscale | 2.93 | bluish_green / reddish_purple | 26.21 | 20.88 | 5.79 |

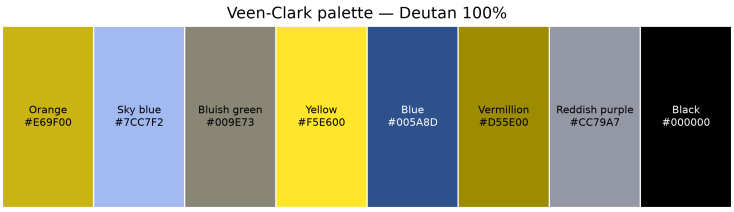
# Palette strips



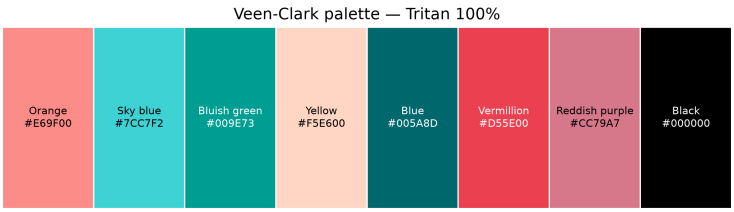

Normal sRGB



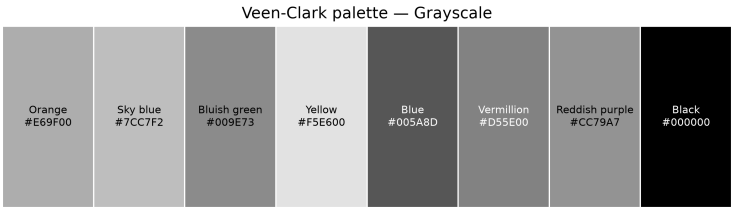

Simulated protan 100%




Simulated deutan 100%




Simulated tritan 100%




WCAG-luminance grayscale

# Severity progressions

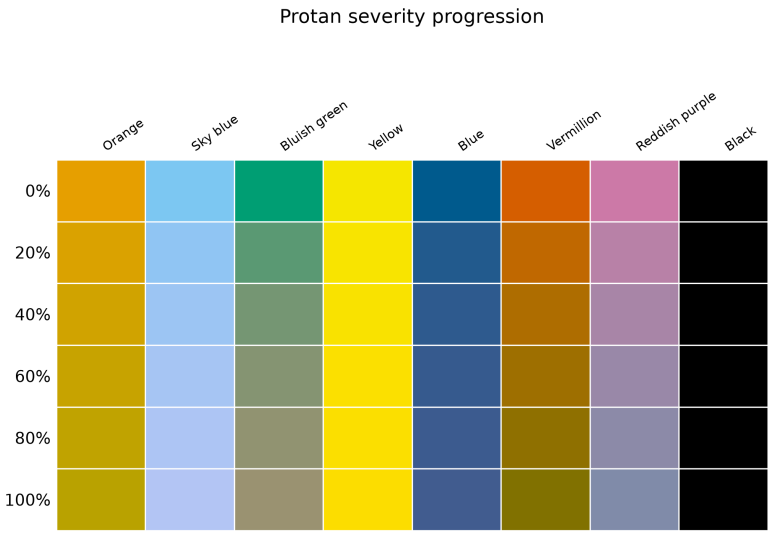



Figure 6. Protan simulation progression from 20% to 100%.

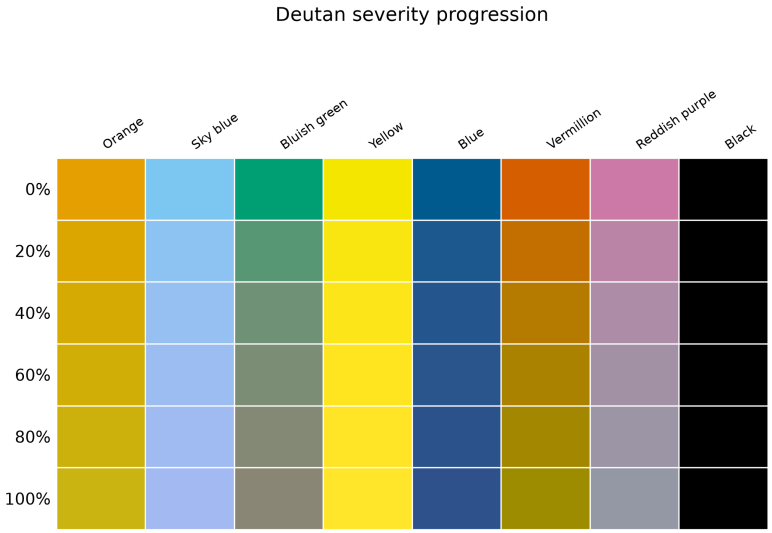



Figure 7. Deutan simulation progression from 20% to 100%.

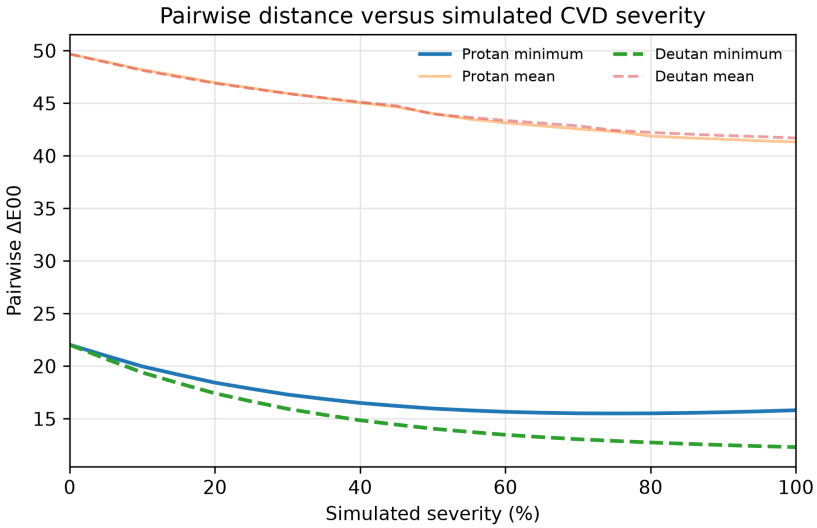
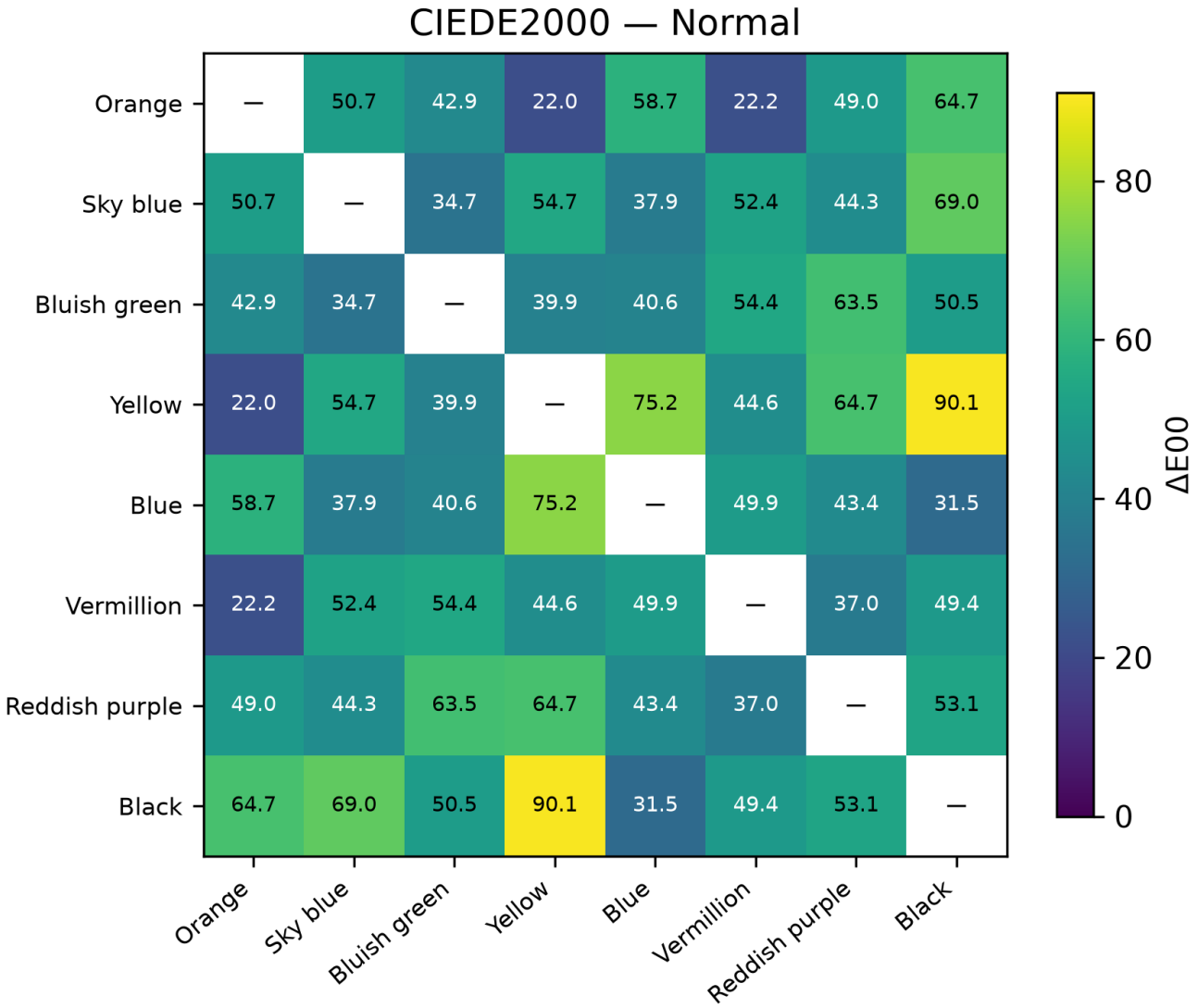


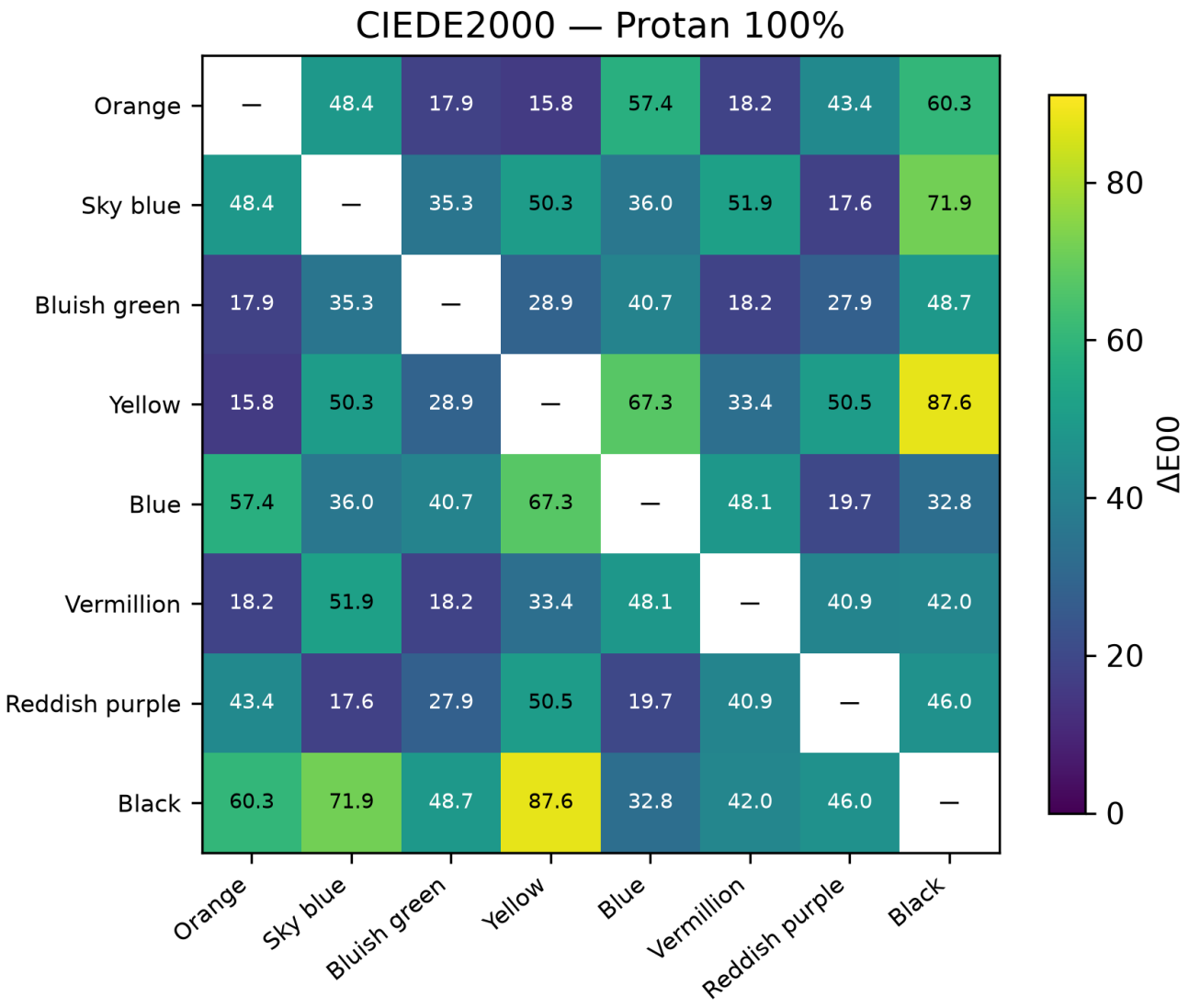

Figure 8. Minimum pairwise distance across simulated severity.

# Normal pairwise heatmap



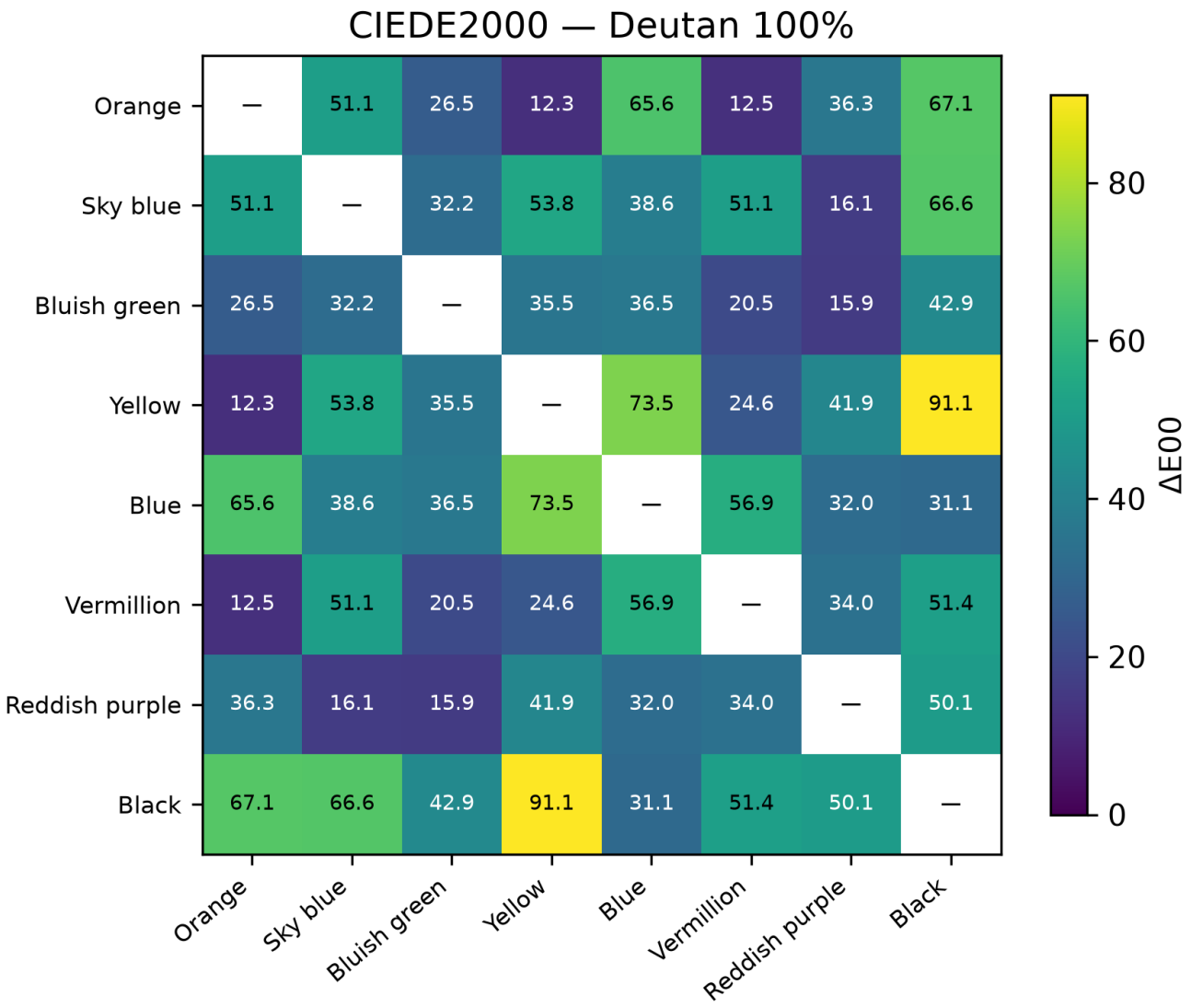

Normal pairwise heatmap.

# Protan 100% pairwise heatmap



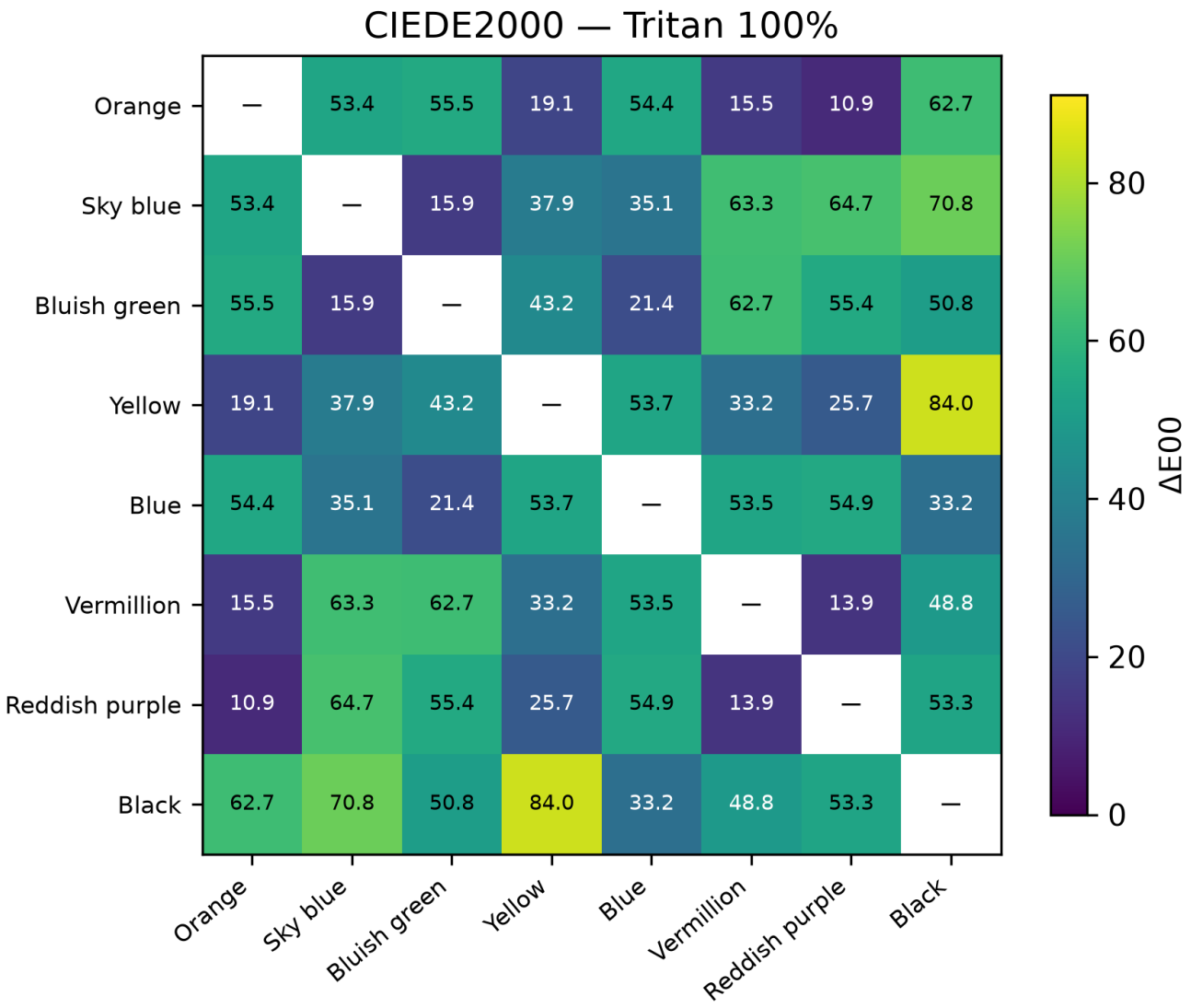

Protan 100% pairwise heatmap.

# Deutan 100% pairwise heatmap




Deutan 100% pairwise heatmap.

# Tritan 100% pairwise heatmap




Tritan 100% pairwise heatmap.

# Normal diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.




Normal diagonal split pair matrix.

# Protan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.




Protan 100% diagonal split pair matrix.

# Deutan 100% diagonal split pair matrix

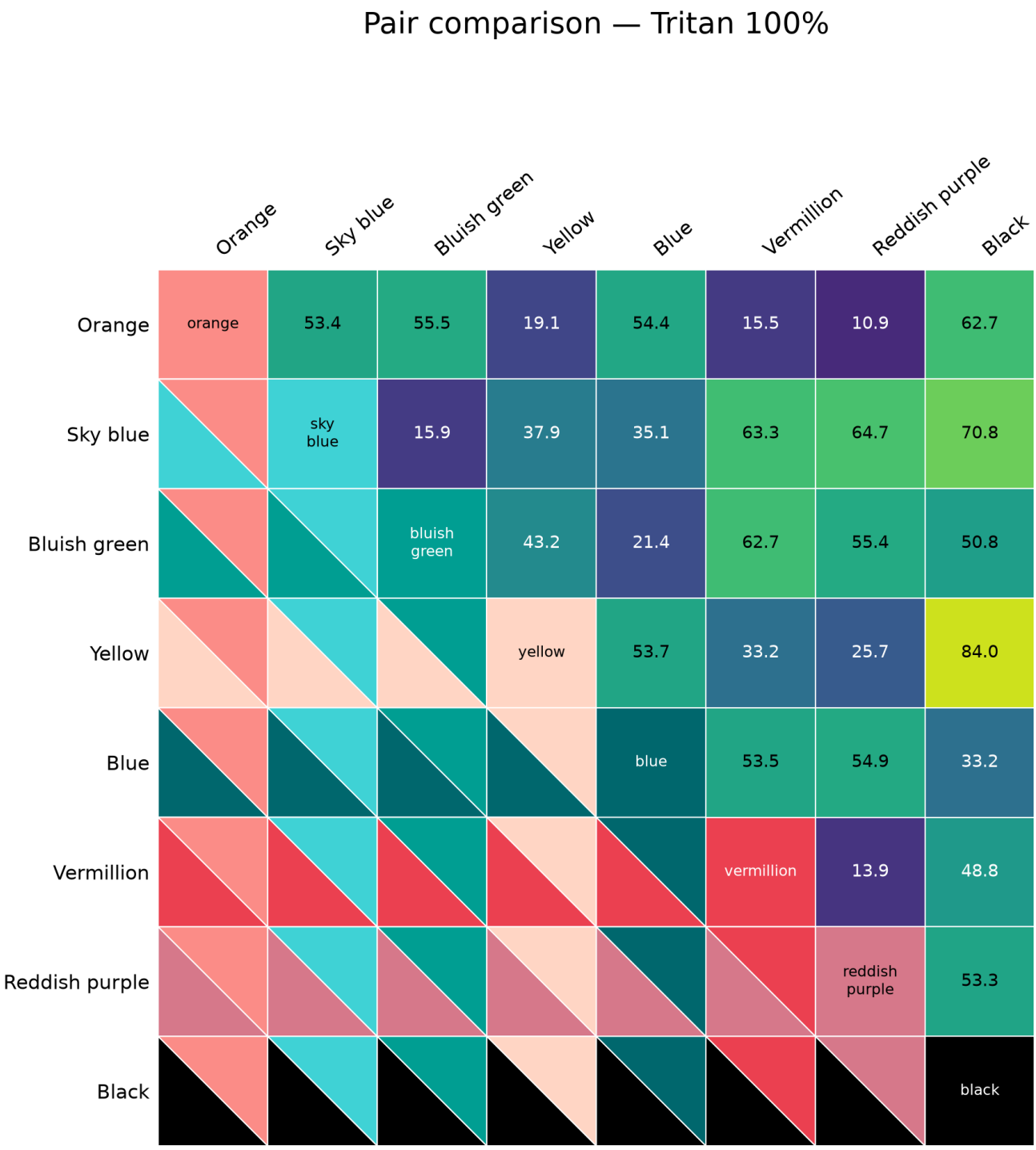
The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.




Deutan 100% diagonal split pair matrix.

# Tritan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.




Tritan 100% diagonal split pair matrix.

# Grouped blue-family analysis

The YAML declares sky blue and blue as one visual family. With two members, the group contains one colour pair: sky blue and blue.

| Condition | Severity | Pairs | Minimum | Mean | Maximum |
|---|---|---|---|---|---|
| normal | 0 | 1 | 37.92 | 37.92 | 37.92 |
| protan20 | 20 | 1 | 37.32 | 37.32 | 37.32 |
| protan40 | 40 | 1 | 36.87 | 36.87 | 36.87 |
| protan60 | 60 | 1 | 36.53 | 36.53 | 36.53 |
| protan80 | 80 | 1 | 36.26 | 36.26 | 36.26 |
| protan100 | 100 | 1 | 36.03 | 36.03 | 36.03 |
| deutan20 | 20 | 1 | 37.83 | 37.83 | 37.83 |
| deutan40 | 40 | 1 | 37.92 | 37.92 | 37.92 |
| deutan60 | 60 | 1 | 38.10 | 38.10 | 38.10 |
| deutan80 | 80 | 1 | 38.32 | 38.32 | 38.32 |
| deutan100 | 100 | 1 | 38.57 | 38.57 | 38.57 |
| tritan100 | 100 | 1 | 35.08 | 35.08 | 35.08 |
| grayscale | 0 | 1 | 37.43 | 37.43 | 37.43 |

# Weakest-pair diagnostic




Figure 9. Weakest pairs under each standard condition.

# Interpretation and limitations

The palette improves its intended blue-family separation in all standard PaletteBench conditions. It does not improve every pair. In particular, the darker blue approaches black, and its tritan-simulated relationship with vermillion loses separation relative to Okabe-Ito. Redundant encodings remain appropriate whenever category identity is critical.

CVD simulations and perceptual-distance models do not reproduce a particular person's perception. Monitor calibration, viewing environment, acuity, ageing, print reproduction, surrounding colours, spatial context, and mark size can all change discriminability. Human evaluation is outside this computational audit.

## Reproducibility

Palette input SHA-256: c0dea2592770ce1867ffdde6c9d169d09e60c531ba0a94994ca0031d69ac88ee. PaletteBench version: 0.1.0. Metric: CIEDE2000 via colour-science; sRGB D65 to CIE Lab. Simulation: Colorspacious sRGB1+CVD (Machado et al.) with display-bound clipping.

# Responsible use of AI

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Generative artificial intelligence (AI) tools, such as ChatGPT and Codex, may be employed as **assistants** in software development and documentation. Their contributions may include implementing human-defined designs, suggesting or refactoring code, debugging, developing tests, improving documentation, exploring technical alternatives, and providing critical feedback. AI-generated suggestions are treated as provisional and are subject to human review, modification, and appropriate verification before adoption.

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Where AI tools are employed as part of the scientific methodology itself, rather than merely as development assistants, their role is documented separately as appropriate to ensure transparency and reproducibility.

## References

1. Okabe, M., and Ito, K. (2002). How to make figures and presentations that are friendly to color blind people. Original brochure (PDF).

2. Wong, B. (2011). Points of view: Color blindness. *Nature Methods*, **8**(6), 441.

3. Wong, B. (2023). Author Correction: Points of view: Color blindness. *Nature Methods*, **20**(8), 1266.

4. Veen, R. J. (2026). PaletteBench (v0.1.0) [Software]. Zenodo. Source code for v0.1.0.